

Principles & Techniques for Managing Essential Processing

WEEK 8 · LESSONS 15 & 16

In cognitive learning theory, not all mental effort is equal. **Essential processing** refers to the irreducible cognitive work required to understand and internalize core content — the mental activity that simply cannot be removed without losing the meaning itself. Unlike extraneous processing (caused by poor design) or generative processing (deep elaboration), essential processing is the baseline your learners must perform just to engage with the material. Managing it effectively is one of the most important skills in content design and UX writing.

This week, we focus on two foundational principles that directly shape how learners experience and absorb content: the **Personalization Principle** and the **Voice Principle**. Both are rooted in empirical research and offer practical, immediately applicable techniques for anyone creating instructional content, digital interfaces, or any form of user-facing communication.

What Is Essential Processing?

FOUNDATIONAL CONCEPT

The Core Idea

Essential processing is the cognitive load associated with comprehending the core content itself. It is determined by the complexity and amount of material — not by how that material is presented. Even in a perfectly designed learning experience, essential processing remains. It is non-negotiable.

Think of it as the "floor" of cognitive effort. Your job as a content designer is not to eliminate it — that would mean removing the learning — but to **manage it thoughtfully** so learners aren't overwhelmed before they even begin to understand.

Why It Matters for Content Designers

When essential processing is high — because the material is dense, abstract, or unfamiliar — learners have less working memory available for deeper understanding. If your presentation also introduces extraneous load through cluttered design or confusing language, the cognitive system quickly becomes overloaded.

Two primary strategies exist for managing essential processing:

- **Segmenting:** Breaking content into smaller, digestible units
- **Pre-training:** Introducing key concepts or vocabulary before the main content

The Personalization and Voice principles operate on a different lever: they reduce the perceived cognitive distance between the content and the learner, making the essential processing feel more natural and less effortful.

The Personalization Principle

PRINCIPLE 1 OF 2

The Personalization Principle states that **people learn more deeply when words are presented in a conversational style rather than a formal style**. This finding, extensively researched by Richard Mayer and colleagues, challenges the assumption that formal, authoritative language is inherently more effective for teaching.



Personalization Principle: What the Research Says

The Core Finding

In study after study, learners who received content written in first- and second-person ("you," "your," "I," "we") consistently outperformed learners who received the same content in third-person, passive, or impersonal language — on both transfer tests and retention measures. The effect holds across multimedia lessons, text-based instruction, and digital interfaces.

The Cognitive Explanation

Conversational language activates what researchers call a "**social partnership**" **cue**. When learners sense they are in a dialogue rather than receiving a broadcast, they subconsciously engage more deeply. They try harder to make sense of the material because they feel the content is speaking directly to them — triggering more active generative processing on top of the essential load.

The Implication

This doesn't mean dumbing content down or being overly casual. It means choosing direct address over abstraction, active voice over passive, and human warmth over institutional distance. The goal is to **lower the perceived effort of essential processing** by making the learner feel like a participant, not a recipient.

Applying the Personalization Principle

Formal vs. Conversational: Side by Side

Formal Style	Conversational Style
The user is required to submit the form before proceeding.	You'll need to submit the form before moving on.
It is recommended that learners review the preceding module.	Before you dive in, it helps to review the last module.
The system will generate a confirmation upon completion.	Once you're done, you'll see a confirmation right away.
Errors may be encountered during the upload process.	If something goes wrong during upload, here's what to try.

Key Techniques

- **Use second person ("you")** to address the learner or user directly throughout your content
- **Use first person ("I," "we")** when the interface or instructor is speaking — it builds perceived presence
- **Prefer active voice** — it is shorter, clearer, and more personal than passive constructions
- **Contract when appropriate** — "you'll" and "it's" feel natural; "you will" and "it is" feel stiff
- **Avoid nominalization** — turn "the completion of the form" into "completing the form"
- **Acknowledge the learner's perspective** — phrases like "you might be wondering..." signal empathy

 The key constraint: personalization should feel authentic, not performative. Forced casualness ("Hey there, champ!") can backfire and undermine credibility.

The Voice Principle

PRINCIPLE 2 OF 2

The Voice Principle focuses specifically on **how spoken or implied vocal presence affects learning**. It states that people learn better when narration is delivered in a human voice rather than a machine voice — and more broadly, that the perceived humanity of the "voice" behind content shapes how deeply learners engage with it.

While originally studied in the context of audio narration in e-learning, the Voice Principle has profound implications for any content that creates an implied narrator — including written UX copy, chatbot dialogue, error messages, and instructional text. Any time your content has a "voice," the Voice Principle applies.

Voice Principle: Key Dimensions



Human vs. Machine Voice

Research shows learners consistently perform better with human narration compared to synthesized speech — even when the words are identical. The warmth, natural prosody, and subtle variation of a human voice serve as a social cue, encouraging learners to engage as if in conversation. In written content, this translates to prose that "sounds" like a real person is speaking.



Enthusiastic vs. Flat Delivery

A second dimension of the Voice Principle concerns expressive range. Content delivered with appropriate enthusiasm and energy leads to better learning outcomes than the same content delivered in a monotone. In writing, "enthusiasm" manifests as sentence variety, strategic emphasis, and language that signals the writer genuinely cares about the material.



Persona Consistency

A consistent, believable voice persona — whether a course instructor, a brand's UX copy style, or an onboarding assistant — helps learners build a stable mental model of who is speaking to them. Inconsistent voice (formal in one section, slangy in another) disrupts this model and adds cognitive friction on top of the essential processing already required.

Voice Principle in UX Writing & Content Design

Beyond Audio: Voice in Written Interfaces

Even when there is no literal audio, every piece of interface copy has a "voice." Error messages, tooltips, onboarding flows, empty states, confirmation dialogs — all of them imply a speaker. Applying the Voice Principle means asking: *Does this sound like a thoughtful, helpful human being?*

Content designers use **voice and tone guidelines** to operationalize this principle across a product. Voice is the consistent personality; tone is how that personality adapts to context (celebratory on a success screen, calm and clear on an error screen).

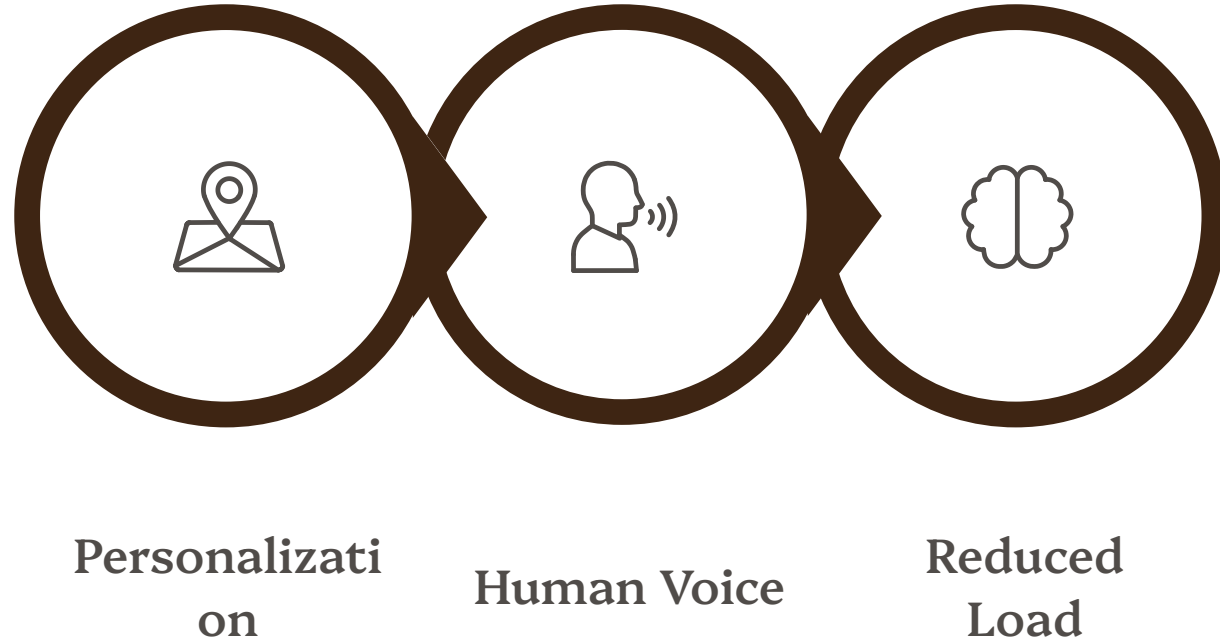
- ✔ A well-defined voice reduces essential processing because learners and users spend less mental energy decoding "who is speaking" — they already know and trust the voice.

Practical Techniques for Voice

- **Write like you talk:** Read your copy aloud. If it sounds stiff or unnatural, revise it.
- **Vary sentence length:** Short punchy sentences create energy. Longer ones build context and nuance. Mixing them creates rhythm.
- **Use conversational transitions:** Words like "Here's the thing," "So," and "That said" signal natural spoken thought patterns.
- **Acknowledge difficulty honestly:** "This next part is a bit technical — stick with us" sounds human. "The following section contains advanced material" does not.
- **Avoid robotic phrasing:** Phrases like "Please be advised that..." or "Kindly note that..." immediately signal a machine or bureaucratic voice, not a helpful human one.
- **Match energy to context:** Celebratory moments deserve enthusiastic copy. Error states deserve calm, reassuring language. The voice adapts — the persona stays consistent.

Personalization & Voice: How They Work Together

SYNTHESIS



While the Personalization Principle and the Voice Principle are studied separately in the research literature, they operate through the same underlying mechanism: **activating social partnership cues** in the learner's mind. When content feels like it is coming from a real, engaged human being speaking directly to you, working memory is deployed more efficiently — the essential processing of understanding core content becomes smoother, less resistant, and ultimately more productive. Together, these principles form the cornerstone of effective conversational content design.

Key Takeaways & What to Apply

WEEK 8 SUMMARY

Essential Processing

It is the irreducible cognitive work required to understand core content. You cannot eliminate it — you can only manage it. Good design lowers the effort required to perform it.

Personalization Principle

Use conversational, direct language — second person, active voice, natural contractions. Learners engage more deeply when they feel spoken to, not lectured at.

Voice Principle

Content with a warm, consistent, human voice reduces cognitive friction. Build a voice persona, vary your sentence rhythm, and match emotional tone to context.

Your Practice Challenge

Take a piece of formal instructional or UX copy you have written recently. Revise it applying both principles. Read it aloud. Notice how the learner relationship shifts.

- 📌 Both principles are backed by robust empirical research. When you apply them intentionally, you are not just "making content friendlier" — you are **optimizing cognitive architecture** to support deeper, more durable learning.